Run report: cover census, run 20261013T0403Z

Descriptive record of one run. Interpretation is in the study `README.md` / `REPORT.pdf`. This run re-executed the same deterministic tool after it gained aggregate `mod4` records and a neighbor-model count; its per-curve records are identical to run `20261013T0305Z` (checked by diff of the `curve` records).

What ran

- Command: `rustc --edition 2021 -O cover_census.rs && ./cover_census 5 && ./cover_census 7 && ./cover_census 11 120`, 8 threads.
- Spec: `PROTOCOL.md` (sha256 in the environment block below), protocol-versioned at commit, not independently preregistered. Fixed seed 20261012 for the q = 11 sample.
- Isolation: none; shared host under heavy load. No timing is a result. All recorded values are exact counts and deterministic.
- Raw records: `results/run-20261013T0403Z.jsonl` (`rec` = field, curve, summary, mod4, totals).

```
run: 20261013T0403Z (supersedes 20261013T0305Z; adds mod4 aggregate records and neighbor-model counts;
curve records must match the earlier run)
command: rustc --edition 2021 -O cover_census.rs && ./cover_census 5 && ./cover_census 7 && ./cover_census
11 120   (8 threads default)
commit (base): 87601f28254ce7d067a54ecbb9ff53d625c8cadf
source sha256: 26f6997d62cf17555e6b19a781c516ccc6b7afc62cef6c720428ef187433e775
protocol sha256 (PROTOCOL.md incl. 2026-10-13 amendment):
8c87bfda479a3cbaca8a98ca1d08e52146134644b9ca2b4506ce64ea5ec76db7
preregistration: protocol-versioned at commit, not independently preregistered (see PROTOCOL.md amendment)
rustc: rustc 1.93.1 (01f6ddf75 2026-02-11) (Homebrew)
host: Darwin 25.6.0 arm64, Apple M4 Pro, 14 cpus
uptime at start: 21:04  up 1 day, 20:43, 8 users, load averages: 98.29 101.47 105.81
q=5 wall_s 2
q=7 wall_s 47
q=11 120 wall_s 284
uptime at end: 21:09  up 1 day, 20:48, 8 users, load averages: 96.28 98.83 102.96
DONE
amendment (2026-10-13): the q=11 arm of the earlier run 20261013T0305Z first failed on a shell invocation
error (arguments passed as one string, no output). The exact malformed command, its exit status and its
stderr were not captured at the time and are unrecoverable: command UNAVAILABLE, exit status UNAVAILABLE,
stderr UNAVAILABLE. The only surviving record is the one-line cause in results/run-20261013T0305Z.env.txt.
This run (0403Z) had no failed attempts.
```

Runs requested and completed

Arm	Requested	Completed	Notes
q = 5	all classes, all structures	252 curves x 16275 structures	
q = 7	all classes, all structures	690 curves x 120099 structures	
q = 11	120 sampled curves and their 2-isogeny neighbours, all structures	120 primary + 127 neighbor models = 247 models, each over 1786323 structures	the earlier run's first attempt failed on a shell quoting error with no output; its command and stderr were not kept (see environment block)

Recorded values

Unit: number of isomorphism classes of curves (q = 5, 7) or sampled curves (q = 11). Counts are exact; no uncertainty is attached. `mismatches` (genus formula against the sum over quadratic subcovers) is 0 in all three arms. The plotted fractions and the n under each group are the tool's `mod4` records, not computed by the plotting script.

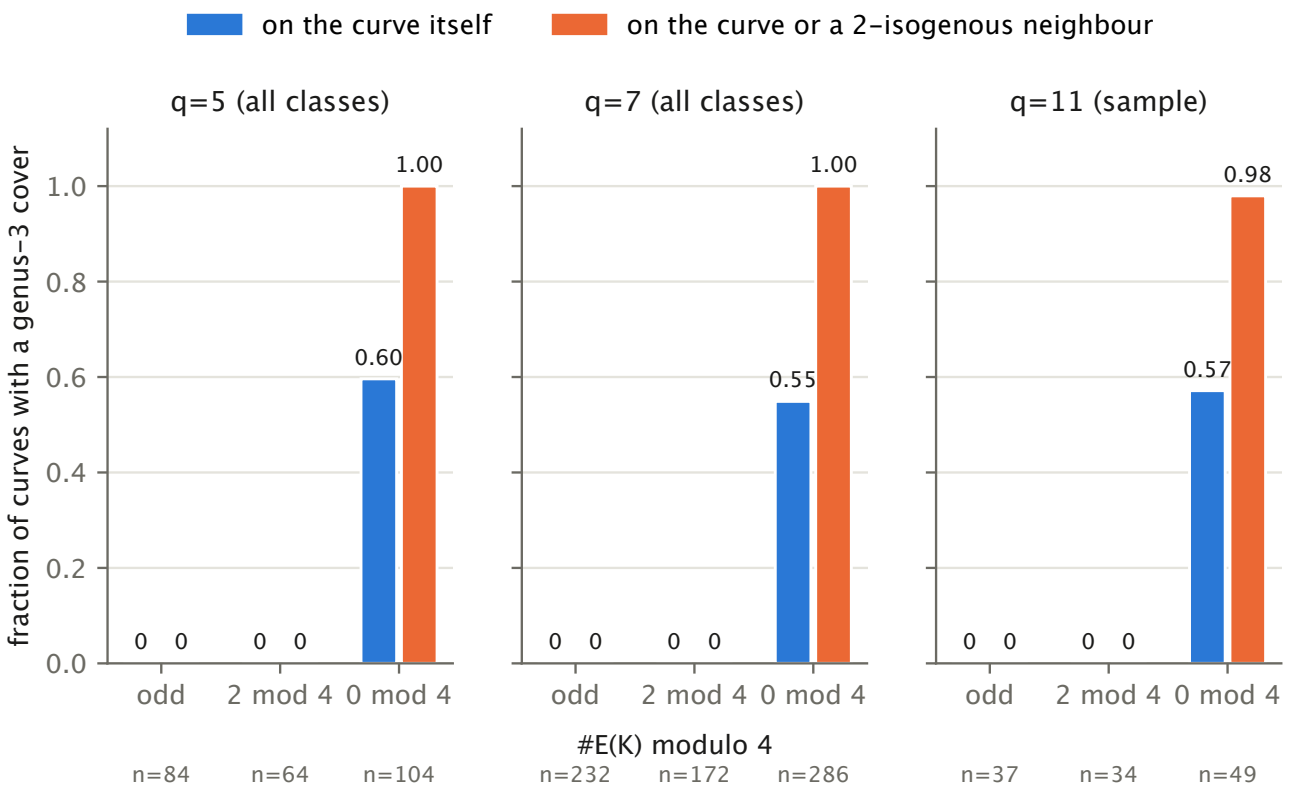


Figure 1: Run 20261013T0403Z. Fraction of curves whose minimal cover genus is 3, by #E(K) mod 4, as recorded in the `mod4` records (`genus3_fraction`); n under each group is the recorded curve count. q = 11 is a sample of 120.

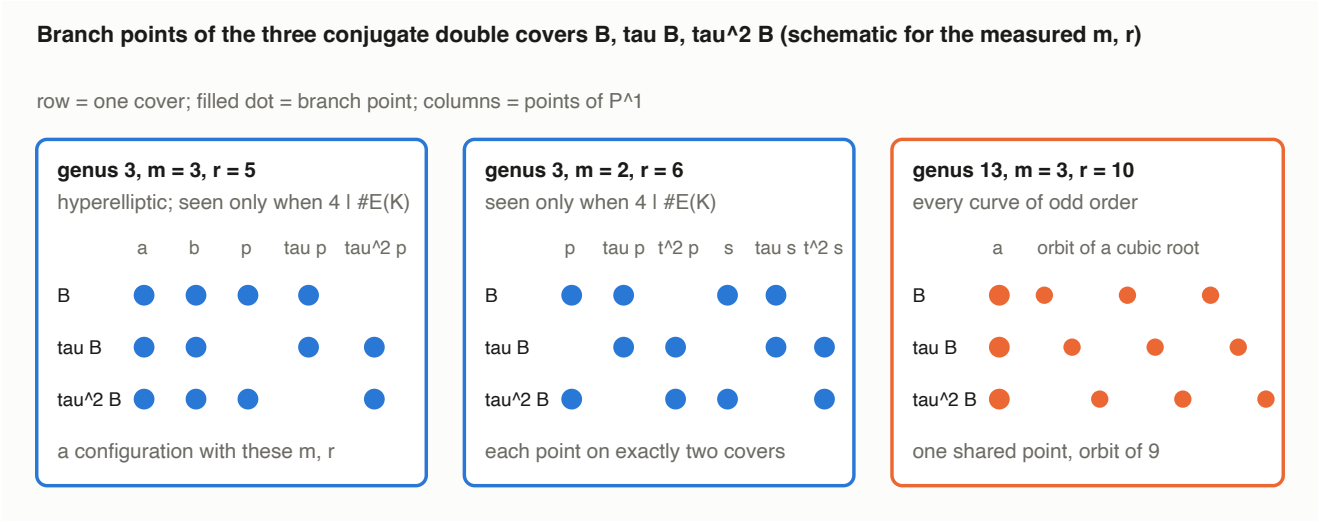


Figure 2: Schematic of the branch configurations behind the measured (m, r) types.

Missing data

- Isogeny-class minimum at q = 11 (sample only).
- Hyperellipticity of the m = 2 genus-3 covers: not decided by branch data.